Hydraulic Press HP-50 Maintenance Manual

50 kN assembly press

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Brandt Hydraulik (fictional)

Read this manual before installing, operating or servicing the equipment. Keep it at the machine.

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1 Safety

The HP-50 hydraulic press is used to press bearings, bushes and housings in assembly lines. Only trained personnel may operate it. Never bridge the two-hand control or the light curtain.

- Hydraulic oil can be hot (up to 60 °C) and under pressure (up to 210 bar).
- Before any work, lower the ram onto blocks, switch off the pump and release the pressure at the test point.
- Wear safety glasses and gloves when working on hydraulics.

2 Description and technical data

Item	Value
Press force	50 kN
Pump motor	18.5 kW, 1,480 rpm, 400 V
Rated current	32 A
Pump	Axial piston, 45 cm³/rev (HYD-PMP-25)
Operating pressure	180–210 bar
Tank	160 l, HLP 46
Motor bearings	2 x SKF-6308
Coupling	Jaw coupling CPL-42
Oil cooler	Air/oil, thermostat 45 °C
Normal vibration RMS (pump unit)	2.2–3.0 mm/s
Normal bearing temperature	48–56 °C
Normal oil temperature	40–50 °C

3 Operation

Start the pump 5 minutes before production so the oil reaches 35 °C. The press cycle is set in the line control. Oil temperature above 60 °C triggers a warning, above 70 °C the pump stops.

4 Hydraulic system

4.1 Oil and filters

Change the return filter FLT-HYD-10 every 1,000 hours or when the clogging indicator turns red. Change the oil (HLP 46, OIL-HLP46-20L) every 4,000 hours or after an oil analysis says so.

4.2 Oil cooler

A fouled oil cooler is the most common cause of overheating. Clean the fins every month with compressed air from the clean side. Check that the fan runs when the oil is above 45 °C.

4.3 Leaks

Check hoses and fittings weekly. Replace any hose that is wet, cracked or older than 6 years.

5 Pump unit

5.1 Coupling alignment

Misalignment between motor and pump raises vibration at 1x and 2x running speed with normal kurtosis and wears the coupling insert (CPL-42). Align with the laser tool: offset max. 0.05 mm, angle max. 0.05 mm/100 mm.

5.2 Motor bearings

The pump motor runs on two 6308 bearings (SKF-6308). Rising kurtosis (above 4) with rising vibration and bearing temperature means bearing damage. Replace both bearings as a pair.

6 Maintenance plan

Interval	Work
Daily	Oil level, leaks, noise
Weekly	Hoses, filter indicator
Monthly	Clean oil cooler
1,000 h	Return filter
4,000 h or 6 months	Oil change, coupling alignment, motor bearing check

6.1 Overheating

Oil and pump motor temperature rise over hours or days while vibration stays almost unchanged. Causes: fouled cooler, broken fan, wrong oil, internal leakage. Check the cooler first.

6.2 Troubleshooting

Symptom	Cause	Action
Oil hot, vibration normal	Cooler fouled, fan	4.2
Vibration rising, kurtosis normal	Coupling misalignment	5.1
Vibration rising, kurtosis above 4	Motor bearing damage	5.2
Slow ram	Pump wear, low pressure	Check pressure
Noise like gravel	Cavitation	Oil level, suction filter